Gell-Mann–Nishijima from the Galois Structure:

$Q = I_3 + Y/2$ algebraically derived

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Abstract

The Gell-Mann–Nishijima relation $Q = I_3 + Y/2$ is derived from two established Galois properties: (1) Hypercharge $Y = -1 + \frac{4}{3}\mathrm{NQR}_{13}$ from the Legendre symbol mod 13 (Doc. 346, Theorem A). (2) Isospin $I_3 = \pm\frac{1}{2}$ from the Su/Sd split via the jE7 projector (Doc. 344, Theorem E). The combination gives all four fermion charges $\{0, -1, +\frac{2}{3}, -\frac{1}{3}\}$ exactly. Theorem E [S] of Doc. 346 is closed: **[B]**. See German version for full proofs.

.1 What remains open: SU$(2)_L$ and CKM [S]

SU$(2)_L$ chirality. Chirality requires $\gamma_5 = e_1 \cdots e_6 \in \mathrm{Cl}(6)$. The $\mathbb{Z}_2$ parity in GF$(27)^*$ separates particles from anti-particles ($k < 13$ vs. $k \geq 13$), not left-handed from right-handed. All four primitive orbits $O_1 \ldots O_4$ are same-chiral. What is needed: the bridge GF$(27)^* \to \mathrm{Cl}(6) \xrightarrow{\gamma_5}$ chirality. The embedding of Galois orbits into the Clifford grade structure is begun in Docs. 341/344 but not yet complete. **[S]**

CKM mixing angles. GF(27) carries a trace bilinear form $\langle a, b\rangle = \mathrm{Tr}_{\mathrm{GF}(27)/\mathrm{GF}(3)}(ab) = ab + (ab)^3 + (ab)^9$. The Cabibbo angle between generations would be the overlap $|\langle O_{2,\mathrm{Gen.1}}, O_{4,\mathrm{Gen.2}}\rangle|/(\|O_2\|\|O_4\|)$. Experimental: $\sin\theta_C \approx 0.225$; Galois candidate $\sin(\pi/14) \approx 0.222$ (1.2 %). Full derivation requires: (1) explicit trace form computation for all orbit pairs; (2) generation assignment within orbits; (3) CKM matrix evaluation. A concrete, executable next step requiring its own document. **[S]**

Bibliography

[1] J. Pascher, *Doc. 344*, FFGFT corpus (Sept. 2026).

[2] J. Pascher, *Doc. 346*, FFGFT corpus (Sept. 2026).